

12.82

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Question:

let $\mathbf{M}$ be a 3×3 symmetric matrix with eigenvalues 1, 2 and 3. let $\mathbf{N}$ be any 3×3 matrix with real eigenvalues such that $\mathbf{MN} + \mathbf{NM} = 3\mathbf{I}$, where $\mathbf{I}$ is a 3×3 identity matrix. Then which of the following cannot be eigenvalue(s) of the matrix $\mathbf{N}$? (ST 2022)

a) $\frac{1}{4}$

b) $\frac{3}{4}$

c) $\frac{1}{2}$

d) $\frac{7}{4}$

Solution:

let $\lambda_1, \lambda_2, \lambda_3$ be the eigenvalues of $\mathbf{N}$
By Eigenvalue decomposition

$$\mathbf{M} = \mathbf{Q}\mathbf{D}\mathbf{Q}^T \quad (1)$$

$$\mathbf{N} = \mathbf{Q}\mathbf{N}'\mathbf{Q}^T \quad (2)$$

where

$$\mathbf{Q}^T\mathbf{Q} = \mathbf{I} \quad (3)$$

$$\mathbf{D} = \text{diag}(1, 2, 3) \quad (4)$$

$$\mathbf{N}' = \text{diag}(\lambda_1, \lambda_2, \lambda_3) \quad (5)$$

$$\mathbf{MN} + \mathbf{NM} = 3\mathbf{I} \quad (6)$$

From (1) and (2) (6) can be simplified as

$$\mathbf{Q}\mathbf{D}\mathbf{N}'\mathbf{Q}^T + \mathbf{Q}\mathbf{N}'\mathbf{D}\mathbf{Q} = 3\mathbf{I} \quad (7)$$

$$\Rightarrow \mathbf{Q}^T(\mathbf{Q}\mathbf{D}\mathbf{N}'\mathbf{Q}^T + \mathbf{Q}\mathbf{N}'\mathbf{D}\mathbf{Q}) = \mathbf{Q}^T(3\mathbf{I}) \quad (8)$$

$$\Rightarrow (\mathbf{Q}^T\mathbf{Q})\mathbf{D}\mathbf{N}'\mathbf{Q}^T + (\mathbf{Q}^T\mathbf{Q})\mathbf{N}'\mathbf{D}\mathbf{Q} = 3\mathbf{Q}^T \quad (9)$$

$$\Rightarrow \mathbf{D}\mathbf{N}'\mathbf{Q}^T + \mathbf{N}'\mathbf{D}\mathbf{Q} = 3\mathbf{Q}^T \quad (10)$$

$$\Rightarrow (\mathbf{D}\mathbf{N}'\mathbf{Q}^T + \mathbf{N}'\mathbf{D}\mathbf{Q})\mathbf{Q} = 3\mathbf{Q}^T\mathbf{Q} \quad (11)$$

$$\Rightarrow \mathbf{D}\mathbf{N}' + \mathbf{N}'\mathbf{D} = 3\mathbf{I} \quad (12)$$

From (4) and (5)

$$\text{diag}(\lambda_1, 2\lambda_2, 3\lambda_3) + \text{diag}(\lambda_1, 2\lambda_2, 3\lambda_3) = 3\mathbf{I} \quad (13)$$

$$\Rightarrow \text{diag}(\lambda_1, 2\lambda_2, 3\lambda_3) = \frac{3}{2}\mathbf{I} \quad (14)$$

$$\Rightarrow \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & 2\lambda_2 & 0 \\ 0 & 0 & 3\lambda_3 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{3}{2} \end{pmatrix} \quad (15)$$

$$\Rightarrow \lambda_1 = \frac{3}{2} \text{ and } \lambda_2 = \frac{3}{4} \text{ and } \lambda_3 = \frac{1}{2} \quad (16)$$

The eigen values of $\mathbf{N}$ are $\frac{3}{2}$ and $\frac{3}{4}$ and $\frac{1}{2}$